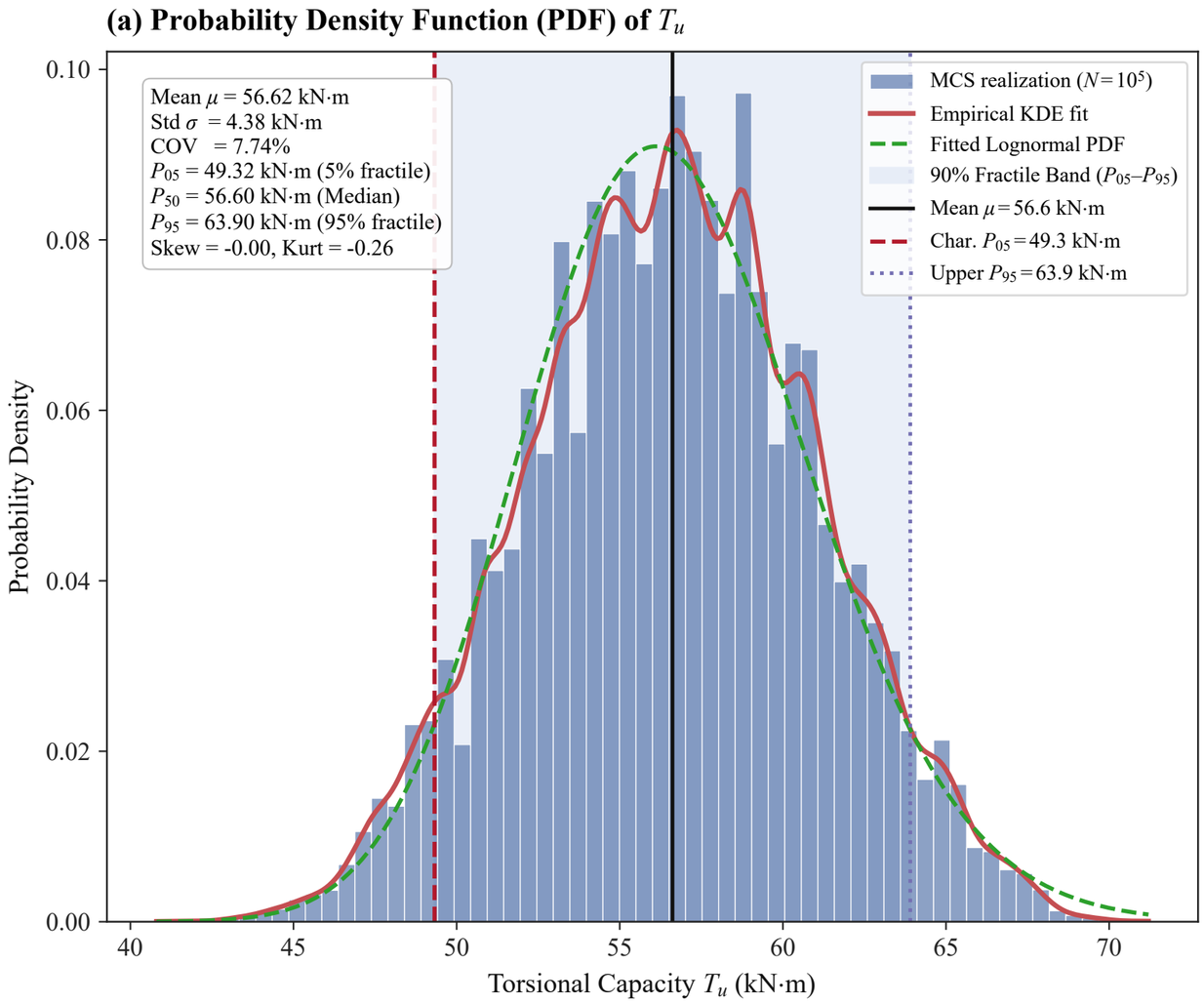
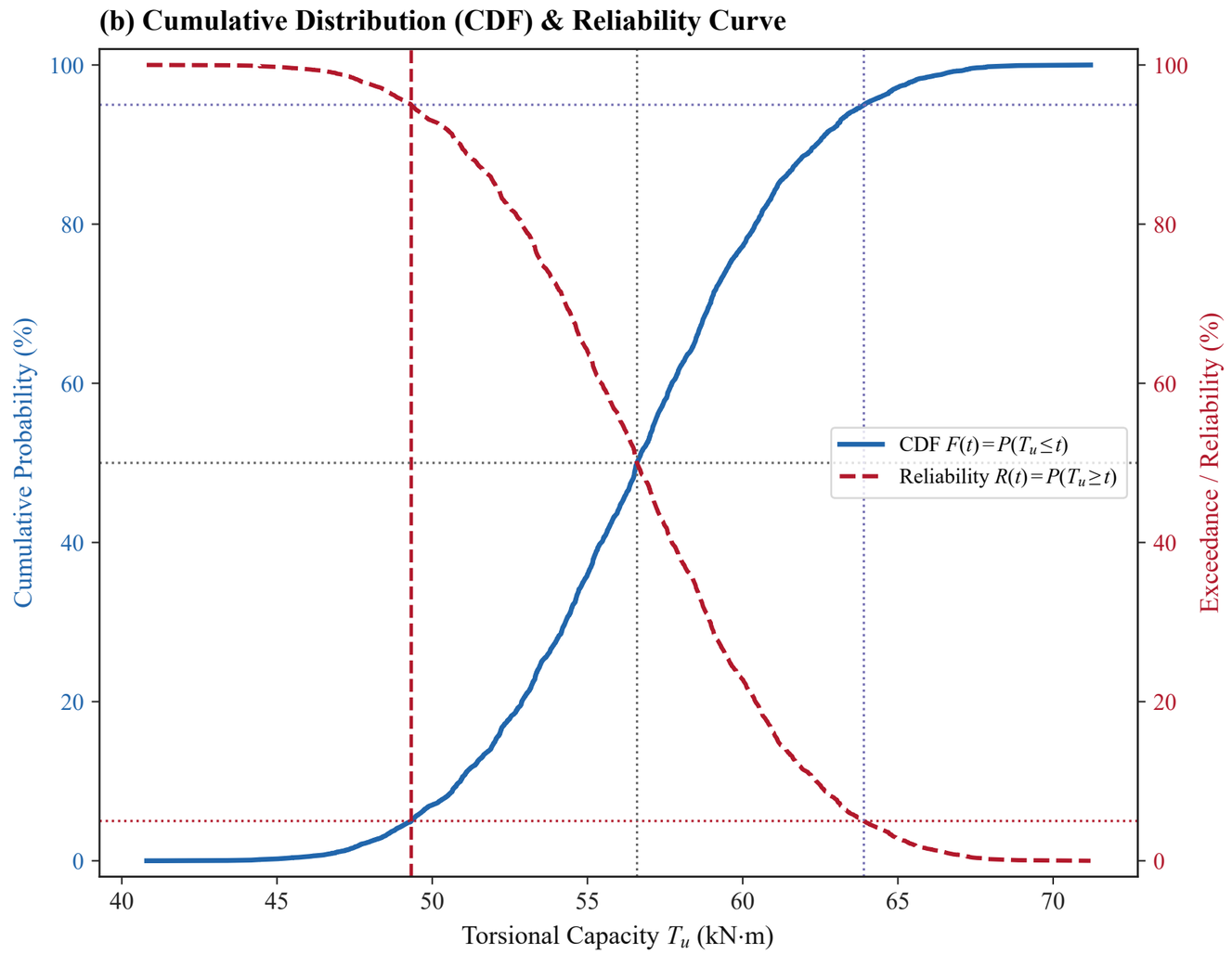
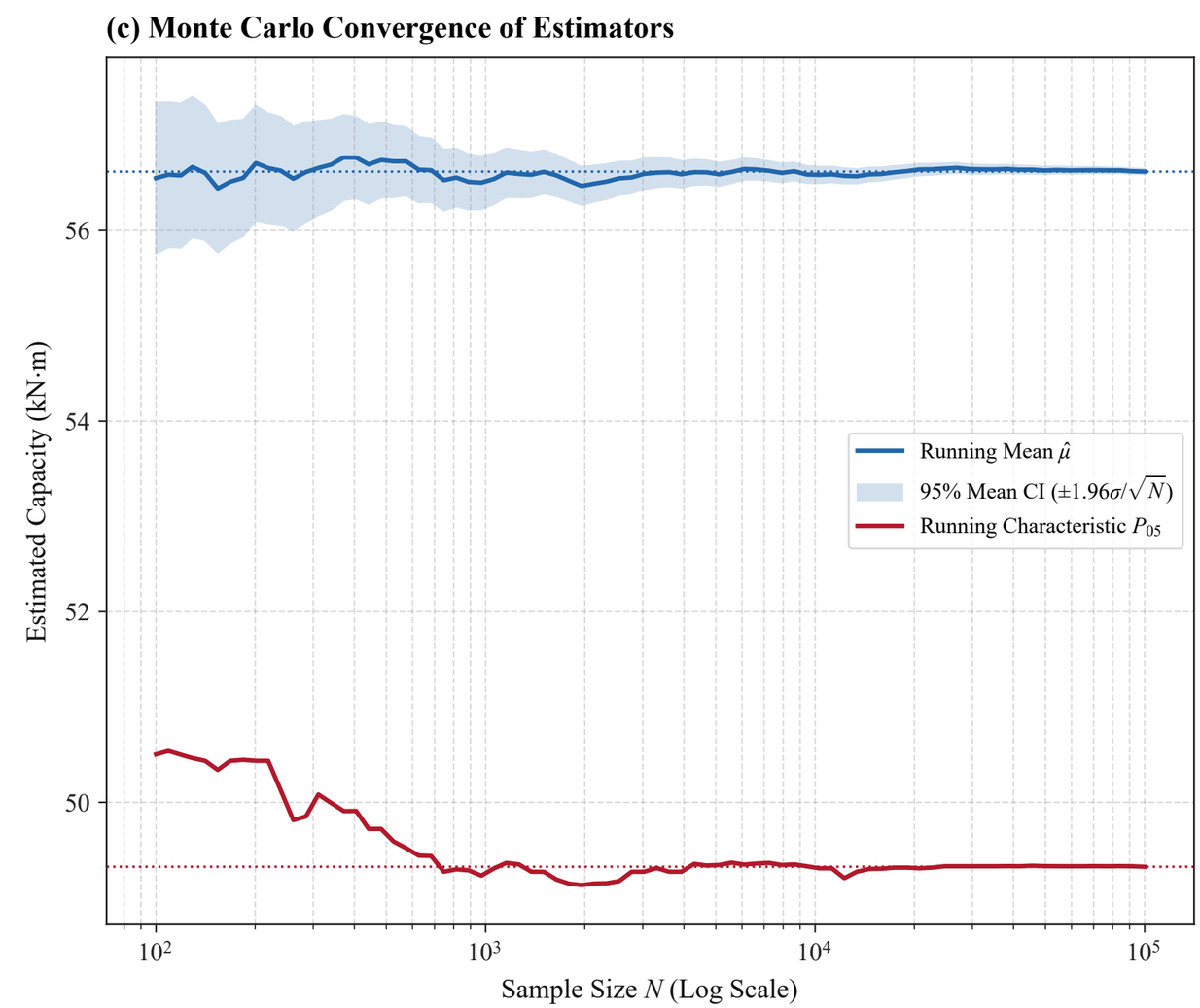
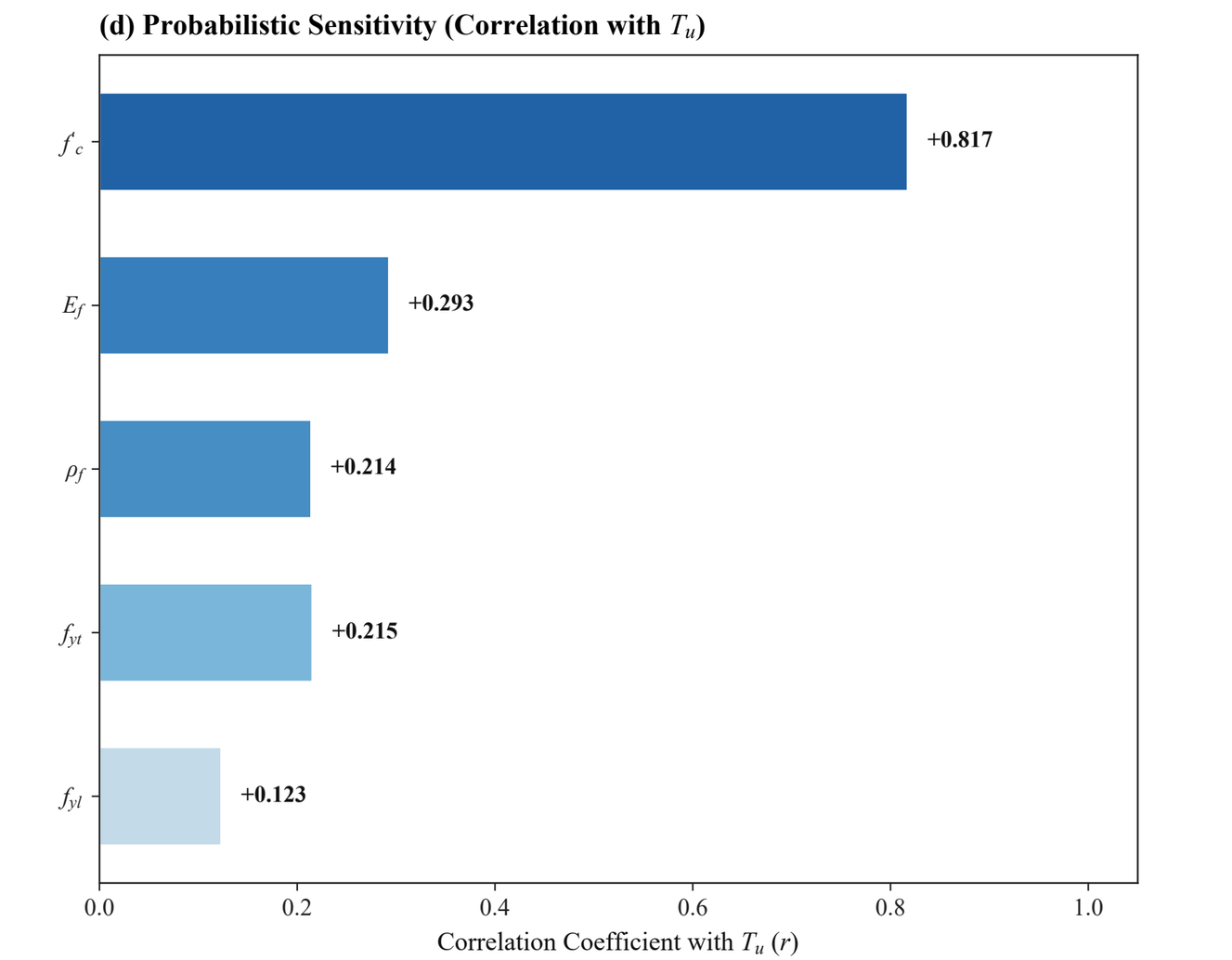

(a) Probability Density Function (PDF) of T_u
Mean μ = 56.62 kN·m
Std σ = 4.38 kN·m
COV = 7.74%
P_{05} = 49.32 kN·m (5% fractile)
P_{50} = 56.60 kN·m (Median)
P_{95} = 63.90 kN·m (95% fractile)
Skew = -0.00, Kurt = -0.26
MCS realization ($N=10^5$)
Empirical KDE fit
Fitted Lognormal PDF
90% Fractile Band (P_{05}–P_{95})
Mean μ = 56.6 kN·m
Char. P_{05} = 49.3 kN·m
Upper P_{95} = 63.9 kN·m
Probability Density
Torsional Capacity T_u (kN·m)


(b) Cumulative Distribution (CDF) & Reliability Curve
CDF $F(t)=P(T_u \le t)$
Reliability $R(t)=P(T_u \ge t)$
Cumulative Probability (%)
Exceedance / Reliability (%)
Torsional Capacity T_u (kN·m)


(c) Monte Carlo Convergence of Estimators
Running Mean $\hat{\mu}$
95% Mean CI ($\pm 1.96\sigma/\sqrt{N}$)
Running Characteristic P_{05}
Estimated Capacity (kN·m)
Sample Size N (Log Scale)


(d) Probabilistic Sensitivity (Correlation with T_u)
f'_c
+0.817
E_f
+0.293
ρ_f
+0.214
f_{yt}
+0.215
f_{yl}
+0.123
Correlation Coefficient with T_u (r)